# E-In Son

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## Research Interests

My research focuses on learning-based navigation for robots in real-world environments, spanning traversability estimation in unstructured terrain, visual navigation, and world models. Going forward, I aim to develop world models and navigation policies that generalize across environments and robot embodiments.

## Education

**Seoul National University** **Mar 2021 – Aug 2027 (Expected)**
*Ph.D. Student, Dept. of Electrical & Computer Engineering* *Seoul, Korea*
Vehicle Intelligence Laboratory. (Advisor: Prof. Seung-Woo Seo)

**Konkuk University** **Mar 2015 – Feb 2021**
*B.S., Electrical & Electronics Engineering* *Seoul, Korea*
*Summa Cum Laude* (GPA 4.33 / 4.5).

## Publications

[1] **Feeling Terrain Before Crossing: World Models for Off-Road Navigation.**
E-In Son, Dong-Wook Kim, Ji-Hoon Hwang, Kangsun Lee, Jisung Bae, Jung-Taak Kim, Seung-Woo Seo.
*Under review*, 2026. [Paper] [Code]

[2] **Which Terrain Is Better? Preference Learning with VLM Prototypes for Off-Road Traversability Ranking.**
Ji-Hoon Hwang, Jisung Bae, E-In Son, Dong-Wook Kim, Jung-Taak Kim, Seung-Woo Seo.
*Under review*, 2026. [Paper]

[3] **CanonNav: Disentangling Navigation Behavior from Camera Geometry in Cross-Platform Visual Navigation.**
Dong-Wook Kim, Ji-Hoon Hwang, E-In Son, Mintaek Oh, Seung-Woo Seo.
*Under review*, 2026. [Paper] [Project] [Code]

[4] **Ordinal Neural Collapse as a Representation Prior for Visual Navigation.**
E-In Son*, Jung-Taak Kim*, Seung-Woo Seo.
*European Conference on Computer Vision (ECCV)*, 2026. [Paper] [Project]

[5] **PTS-Map: Probabilistic Terrain State Map for Uncertainty-Aware Traversability Mapping in Unstructured Environments.**
Dong-Wook Kim, E-In Son, Chan Kim, Ji-Hoon Hwang, Seung-Woo Seo.
*IEEE Robotics and Automation Letters (RA-L)*, 2025. [Paper] [Code]

[6] **Follow the Footprints: Self-supervised Traversability Estimation for Off-road Vehicle Navigation based on Geometric and Visual Cues.**
Yurim Jeon, E-In Son, Seung-Woo Seo.
*IEEE International Conference on Robotics and Automation (ICRA)*, 2024. [Paper] [Code]

[7] **Adaptive Robot Traversability Estimation Based on Self-Supervised Online Continual Learning in Unstructured Environments.**
Hyung-Suk Yoon, Ji-Hoon Hwang, Chan Kim, E-In Son, Se-Wook Yoo, Seung-Woo Seo.
*IEEE Robotics and Automation Letters (RA-L)*, 2024. [Paper]

[8] **Traversability-aware Adaptive Optimization for Path Planning and Control in Mountainous Terrain.**
Se-Wook Yoo*, E-In Son*, Seung-Woo Seo.
*IEEE Robotics and Automation Letters (RA-L)*, 2024. [Paper] [Project]

## Awards & Honors

**Earth Rover Challenge (1st Place among AI Teams)** Oct 2024
*IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*
Led planning for our team's navigation system, deployed remotely on sidewalk robots in 8 cities worldwide and benchmarked against human teleoperators.

**Academic Excellence Scholarship** 2018 – 2020
*Konkuk University*

## Research Projects

**Challengeable Future Defense Technology R&D Program (Phase 2)** 2024 – 2026
*Agency for Defense Development (ADD)*

- Led planning for an end-to-end visual navigation system in unknown, hazardous terrain, building on our visual navigation methods [3, 4].
- Deployed on a Clearpath Husky and a custom tracked robot at 10 Hz, validated over 20 field runs of 1 km each.

**Challengeable Future Defense Technology R&D Program (Phase 1)** 2022 – 2023
*Agency for Defense Development (ADD)*

- Developed traversability-aware path planning and tracking for steep mountainous terrain, adapting to the robot's speed and torque; validated on 27 simulated terrain types and in real-world tests [8].

## Teaching Experience

**Seoul National University**, Teaching Assistant

SNU in Silicon Valley Program, joint with UC Berkeley *(Summer 2024)*
Fundamentals of Random Variables and Probability Processes *(Spring 2023)*
Network Protocol and Design *(Fall 2021, 2023)*

## Academic Service

**Conference Reviewer:** ICRA, IROS, ECCV
**Journal Reviewer:** RA-L

## Skills

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|---|---|
| **Programming** | Python, C/C++, Git |
| **Frameworks** | PyTorch, ROS / ROS 2 |
| **Robotics** | SLAM & localization, semantic segmentation, path planning & tracking, multi-sensor integration (LiDAR, camera, GNSS, IMU) |
| **Simulation** | Isaac Sim, MuJoCo, Gazebo, CARLA |
| **Communication** | Korean (native), English |

## Reference

**Advisor: Prof. Seung-Woo Seo** sseo@snu.ac.kr Website